DATA READINESS SYSTEM (DRS)

MVP SCOPE DOCUMENT (MVP-SCOPE v1.1)

Organization: The AI Institute Africa
System Name: Data Readiness System (DRS)
Document Type: MVP Scope Definition
Version: MVP Scope v1.1
Prepared By: Wills
Date: February 2026
Project Timeline: 18 February 2026 – 11 April 2026
Methodology: V-Model + Hierarchical Model

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1. PURPOSE OF THIS DOCUMENT

This document defines the **official scope of the Data Readiness System MVP**, including:

- What must be implemented
- What is excluded
- Required deliverables
- Minimum system capabilities
- Optional integration scope (external ingestion platform)
- MVP success criteria
- Constraints and assumptions

This scope document serves as the foundation for:

- Sprint planning
- Task assignment
- Implementation decisions
- Stakeholder validation
- MVP acceptance testing

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2. MVP GOAL STATEMENT

The MVP must demonstrate that DRS can:

Ingest evidence documents, preserve them immutably, extract structured candidate data, support human validation, map approved results into a Canonical Data Model (CDM), resolve entities into golden records, and publish traceable datasets with audit logs and governance controls.

The MVP must be deployable in a scalable architecture and be adaptable to any organization without redesign.

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3. MVP SYSTEM BOUNDARIES

3.1 In Scope (MVP Includes)

The MVP includes:

- Evidence ingestion and evidence storage
- Evidence tracking and provenance metadata
- OCR / document intelligence pipeline
- Extraction output storage and confidence tracking
- Human-in-the-loop validation workflow
- Canonical data mapping and schema registry v1.0
- Entity resolution (basic golden record creation)
- Dataset publishing with dataset cards and versioning
- Audit logs and governance controls (RBAC)
- Optional external ingestion platform integration via REST API

3.2 Out of Scope (MVP Excludes)

The MVP will not include:

- Full production-grade connectors for SharePoint/Google Drive
- Advanced AI model training loops
- Advanced anomaly detection and fraud scoring
- Full real-time streaming ingestion
- Advanced knowledge graph reasoning engines
- Full regulatory compliance certification (ISO 27001)
- Multi-region cloud failover
- Full-scale distributed processing optimization
- Automated de-identification/pseudonymization engine (unless explicitly required)

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4. MVP CORE ARCHITECTURE (LAYERS IN SCOPE)

The MVP shall implement all six DRS layers as follows:

4.1 Evidence Layer (MVP Scope)

MVP Capabilities

- Upload documents via UI and REST API
- Generate immutable evidence record with SHA-256 hash
- Store evidence file in object storage
- Maintain evidence metadata registry in database
- Maintain batch and folder grouping support
- Generate evidence lineage reference IDs

Evidence Layer Outputs

- Evidence objects
- Evidence metadata records
- Immutable storage URIs
- Evidence integrity status

Minimum Evidence Data Fields

- evidence_id
- tenant_id
- batch_id
- source_type
- file_hash
- file_uri
- ingested_at
- ingested_by
- document_type_hint
- status

4.2 Intelligence Layer (MVP Scope)

MVP Capabilities

- OCR pipeline execution (basic)
- Document type classification (basic heuristic or ML model)
- Entity extraction (key-value extraction)
- Table extraction (optional, minimum viable support)
- Generate candidate structured outputs with confidence scores

Intelligence Layer Outputs

- OCR text output
- Candidate extracted fields
- Confidence scores per field
- Evidence pointers to extracted text locations

4.3 Canonical Data Layer (MVP Scope)

MVP Capabilities

- Canonical Data Model (CDM v1.0) implementation
- Schema registry (versioned schema)
- Mapping validated extraction fields into CDM entities
- Store canonical entities in database
- Maintain CDM entity lineage back to evidence

Canonical Entities Required (CDM v1.0)

- Document
- Party (Person/Organization)
- Event (Transaction/Action)
- Relationship
- Dataset Version Metadata

4.4 Knowledge Layer (MVP Scope)

MVP Capabilities

- Basic entity resolution (deduplication)
- Golden record creation for parties
- Relationship linking between canonical entities
- Store resolved entities in knowledge store (relational or graph-lite)

MVP Knowledge Outputs

- golden_record_id
- merged entities list
- match score and merge decision logs

4.5 Publishing Layer (MVP Scope)

MVP Capabilities

- Generate dataset exports from canonical records
- Dataset versioning (v1.0.0 format)
- Dataset card generation
- Dataset download interface via UI and API
- Store dataset metadata and lineage

Supported Export Formats (MVP)

- CSV (mandatory)
- JSON (mandatory)
- Parquet (optional)

Dataset Card Minimum Contents

- dataset name and description
- schema version
- dataset version
- lineage evidence batches
- validation summary
- publishing timestamp
- approvals (if required)

4.6 Governance Layer (MVP Scope)

MVP Capabilities

- RBAC enforcement
- Audit logging for critical actions
- Permission controls for publishing/exporting datasets
- Access logs for evidence retrieval and downloads

Mandatory Audit Events

- UPLOAD_EVIDENCE
- OCR_RUN
- EXTRACTION_RUN
- CREATE_VALIDATION_TASK
- APPROVE_FIELD
- REJECT_FIELD
- MAP_TO_CDM
- MERGE_ENTITIES
- PUBLISH_DATASET
- DOWNLOAD_DATASET

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5. MVP SUBSYSTEMS INCLUDED

The MVP must implement the following subsystems:

5.1 Subsystem 1: Physical Operations & Custody (Limited MVP)

MVP Capabilities

- Batch creation and tracking
- Folder ID tagging support (manual entry)
- Custody log record creation (basic)

Exclusions

- QR code printing automation
- full physical storage management system

5.2 Subsystem 2: Ingestion & Immutable Evidence (Full MVP)

MVP Capabilities

- File upload UI
- REST API ingestion endpoint
- Hash generation

- Object storage persistence
- Evidence registry database

5.3 Subsystem 3: Document Intelligence (Core MVP)

MVP Capabilities

- OCR
- extraction pipeline
- confidence scores

5.4 Subsystem 4: Trust & Validation (Full MVP)

MVP Capabilities

- trust score generation
- validation queue UI
- approve/reject fields
- validation workflow states

5.5 Subsystem 5: Canonical Data & Schema Registry (Full MVP)

MVP Capabilities

- CDM mapping service
- schema registry v1.0
- canonical storage tables

5.6 Subsystem 6: Knowledge Sync & Entity Resolution (Basic MVP)

MVP Capabilities

- entity matching rules

- golden record creation

5.7 Subsystem 7: Dataset Publishing (Full MVP)

MVP Capabilities

- dataset generation and export
- dataset cards

5.8 Subsystem 8: Governance & Compliance (Core MVP)

MVP Capabilities

- RBAC
- audit logs
- approval enforcement for publishing

5.9 Subsystem 9: Platform Ops (Basic MVP)

MVP Capabilities

- monitoring hooks (logs + metrics)
- backup support (manual or scripted)
- configuration file templates

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6. OPTIONAL INTEGRATION SCOPE (NEW ADDITION)

6.1 Purpose

The DRS MVP shall support optional integration with an external stand-alone system that provides:

- file ingestion

- file storage
- file tracking
- file lifecycle management

DRS will integrate with this system using REST APIs.

6.2 Integration Principles

The integration must:

- Be optional (DRS must still operate independently)
- Support evidence registration without direct file upload
- Support evidence retrieval via external reference IDs
- Maintain provenance linkage between DRS and external platform
- Not break immutability guarantees

6.3 Integration Modes (MVP)

Mode A: DRS Native Ingestion (Default)

- DRS receives file upload
- DRS stores file in its own evidence store

Mode B: External Platform Ingestion (Optional)

- External platform stores file
- DRS receives metadata + file reference ID
- DRS creates evidence record linked to external platform

6.4 Required External API Contract (Minimum)

The external platform must expose:

API 1: File Metadata Fetch

```
GET /external/files/{file_id}/metadata
```

Returns:

- file_name
- file_hash (optional)
- created_at
- file_type
- storage_uri or download link

API 2: File Download Access

```
GET /external/files/{file_id}/download
```

Returns file stream or signed URL.

API 3: File Tracking Status

```
GET /external/files/{file_id}/status
```

Returns:

- ACTIVE / ARCHIVED / DELETED

6.5 Integration Outputs

When integrated, DRS must store:

- external_platform_id
- external_file_id
- external_tracking_status
- sync_timestamp
- external_uri

6.6 MVP Acceptance Criteria for Integration

Integration is successful if:

- DRS can register external file as evidence
- DRS can retrieve and process the external file
- DRS maintains evidence hash integrity validation

- DRS can trace dataset outputs back to external file reference

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7. MVP USER ROLES IN SCOPE

The MVP must support the following actors:

Actor	MVP Responsibilities
Archive Operator	upload evidence, create batches
Data Validator	validate extracted fields
AI Engineer	configure extraction and CDM mapping
Compliance Officer	approve publishing, review audits
Admin	manage roles, tenants, system configs

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8. MVP USER INTERFACES (UI SCREENS REQUIRED)

The MVP shall include a basic web portal with:

Required Screens

1. Login screen
2. Dashboard overview (counts, statuses)
3. Evidence upload screen
4. Evidence list and search
5. Evidence detail view (file preview + metadata)
6. Validation task queue
7. Validation review screen (approve/reject fields)
8. Canonical record view (CDM records)
9. Dataset publishing screen
10. Dataset list and download
11. Dataset card view
12. Audit log viewer

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9. MVP DATA FLOW (END-TO-END PIPELINE)

The MVP shall support the following end-to-end pipeline:

1. Create batch (optional)
2. Upload or register evidence
3. Evidence stored and hashed
4. OCR job created
5. Extraction job created
6. Trust scoring performed
7. Validation tasks generated
8. Validator approves/rejects
9. CDM mapping executed
10. Entity resolution executed
11. Dataset publishing triggered
12. Dataset card generated
13. Dataset export generated
14. Audit logs available for review

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10. MVP TECHNICAL REQUIREMENTS (MINIMUM)

10.1 Storage

- Object storage for evidence and dataset files
- PostgreSQL for metadata, validation tasks, CDM entities
- Optional graph database not required for MVP

10.2 Processing

- Background job queue required (Celery/RQ/Temporal)

10.3 Security

- JWT authentication required
- RBAC required

10.4 Scalability Requirements

- support 10,000 evidence objects without redesign
- support asynchronous pipeline scaling

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11. MVP DELIVERABLES (MUST BE PRODUCED)

By April 11, the following must be delivered:

Technical Deliverables

- Functional DRS web platform
- Backend services deployed
- Evidence store working
- OCR and extraction pipeline working
- Validation workflow working
- CDM mapping working
- Entity resolution basic working
- Dataset publishing working
- Dataset card generation working
- Audit logs and RBAC working
- External integration module working (optional)

Documentation Deliverables

- MVP scope document (this)
- ICD v1.0
- CDM v1.0
- API documentation (Swagger/OpenAPI)
- Deployment guide
- User guide
- Testing report

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12. MVP SUCCESS CRITERIA (ACCEPTANCE)

The MVP is considered successful if it can demonstrate:

Core Functional Success

✅ Evidence upload and storage
✅ OCR + extraction produces structured output

✅ Validation workflow works end-to-end
✅ Canonical data is generated and stored
✅ Golden records are created (basic)
✅ Dataset publishing works with versioning
✅ Dataset cards generated correctly
✅ Audit logs show traceability
✅ RBAC blocks unauthorized access

Integration Success (Optional)

✅ Evidence can be registered from external ingestion platform
✅ Evidence can be retrieved via external API
✅ Pipeline can process external evidence same as native ingestion

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13. MVP NON-FUNCTIONAL TARGETS

Category	MVP Target
Availability	95% uptime during testing
Response time	< 3 seconds for dashboard queries
Upload handling	support 100MB documents
Evidence integrity	SHA256 must match stored hash
Security	RBAC enforcement on all endpoints
Logging	audit logs stored for all critical actions

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14. ASSUMPTIONS & CONSTRAINTS

Assumptions

- OCR model availability exists (open-source or API)
- Users have role-based access accounts
- Evidence files are not corrupted beyond OCR ability
- External platform provides stable REST APIs

Constraints

- Time: Feb 18 – Apr 11 (8 weeks)
- Team: 3 developers

- MVP must be deployable and demo-ready
- Integration must not compromise immutability

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15. MVP PHASE BREAKDOWN (IMPLEMENTATION VIEW)

Sprint 0 (Architecture + Setup)

- freeze CDM and ICD baseline
- repo setup
- environment setup
- UI skeleton

Sprint 1 (Evidence + Ingestion)

- ingestion module
- evidence storage
- evidence UI

Sprint 2 (Intelligence + Extraction)

- OCR integration
- extraction output storage

Sprint 3 (Validation Workflow)

- validation UI and backend
- trust scoring

Sprint 4 (Canonical Data + Knowledge)

- CDM mapping
- entity resolution basic

Sprint 5 (Publishing + Governance)

- dataset publishing
- dataset cards
- audit logs and RBAC

Sprint 6 (Integration + Stabilization)

- external ingestion platform integration
- bug fixes
- demo preparation

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16. SIGN-OFF

This MVP scope is approved for implementation and will be used for:

- sprint planning
- development task assignment
- system verification
- MVP acceptance testing

Prepared by: Wills
Approved by: _____
Date: _____